

Lecture 10 - DHCPv4 & NAT

Summer 2025 (A)

8. A laptop is assigned an IP address by DHCP with an **8-hour lease**. After **4 hours** of active use, it tries to renew the lease.

I. Which message does the laptop send to attempt renewal?

II. What happens if the server is available and responds?

III. What fallback mechanism occurs if the server does not respond?

I. Message sent to attempt renewal

- The laptop sends a **DHCPREQUEST** message.
- This is typically a **unicast** message sent directly to the DHCP server that originally assigned the IP address.

II. If the server is available and responds

- The DHCP server replies with a **DHCPACK** message.
- The lease is **successfully renewed**, and:
 - The laptop keeps the same IP address.
 - The lease timer is **reset to 8 hours**.
 - Normal network operation continues without interruption.

III. Fallback mechanism if the server does not respond

- The laptop continues using its current IP address.
 - It broadcasts a **DHCPREQUEST** message to **any available DHCP server**.
 - If a server responds with **DHCPACK**, the lease is renewed.
 - If **no server responds before the lease expires**, the client must:
 - Stop using the IP address.
 - Restart the DHCP process with a **DHCPDISCOVER** message
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9. Two PCs in a LAN (192.168.10.10 and 192.168.10.11) connect at the same time to different web servers on the Internet. The ISP gives only one public IPv4 address (209.165.200.225).

I. How can both PCs use the Internet with just one public IP?

II. How does the router know which reply belongs to which PC?

III. What happens if the same port number is already in use?

I. How can both PCs use the Internet with just one public IP?

- The router uses **PAT (also called NAT Overload)**.
- Both private IPs (192.168.10.10 and 192.168.10.11) are translated to the **same public IP 209.165.200.225**.
- The router assigns **different source port numbers** for each outgoing connection.
- This allows multiple internal devices to share one public IPv4 address simultaneously.

II. How does the router know which reply belongs to which PC?

- The router maintains a **NAT translation table**.
- Each entry maps:
 - **Inside private IP + private port**
 - **to public IP + unique public port**
- When a reply arrives, the router checks the **destination port number** and forwards the packet to the correct internal PC.

Example:

- 192.168.10.10 : 49152 → 209.165.200.225 : 30001
- 192.168.10.11 : 49153 → 209.165.200.225 : 30002

III. What happens if the same port number is already in use?

- The router **selects a different available port number**.
- PAT ensures every active connection has a **unique port mapping**.
- If no ports are available (very rare), the new connection request is **dropped**.

Spring 2025(A)

8. When a PC connects to a DHCP-enabled network, it initially broadcasts a message to look for available DHCP servers.

I. **Why does** the PC use a broadcast message instead of a unicast at this stage?

II. **What kind of information** does the PC receive in response, and how does it reply?

I. Why the PC uses a broadcast message

- When the PC first joins the network, it **does not yet have an IP address**.
- Because it does not know:
 - its own IP address, or
 - the IP address of any DHCP server.So, it **cannot send a unicast message**.
- Therefore, the PC sends a **broadcast DHCPDISCOVER** message so that **any available DHCP server** on the local network can receive it.

II. Information received and how the PC replies

- A DHCP server responds with a **DHCPOFFER** message.
- The offer contains:
 - a proposed IP address for the PC,
 - subnet mask,
 - default gateway,
 - DNS server information,

- lease duration.
 - The PC then replies with a **DHCPREQUEST** message to:
 - accept one specific offer, and
 - inform other DHCP servers that their offers are not accepted.
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9.A student project web server is hosted inside a university computer lab at private IP address **10.10.5.50**, listening on port **8080**. The lab is connected to the internet via a router that uses **NAT** with a public IP address. When the student shares the public IP with a recruiter to view the project remotely, the recruiter reports that the link does not work.

I. Why can't the recruiter access the server using the public IP address?

II. What network configuration should be set up on the router to make the server accessible from outside the university?

I. Why the recruiter can't access the server using the public IP

- The web server is hosted on a **private IP address (10.10.5.50)**, which is **not routable on the Internet**.
- The router is using **NAT**, which:
 - allows **outgoing connections** from inside the lab,
 - but **blocks unsolicited incoming connections** by default.
- Since the router has **no rule telling it where to forward incoming requests**, traffic sent to the public IP on port **8080** is dropped.
- As a result, the recruiter's request never reaches the internal web server.

II. Network configuration needed on the router

- The router must be configured with **Port Forwarding**.
- A rule should be added such as:

Public IP : 8080 → 10.10.5.50 : 8080

- This tells the router to:
 - accept incoming traffic on port **8080** of the public IP,
 - forward it to the internal server at **10.10.5.50** on port **8080**.
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Fall 2024(A)

8.A network administrator configures a new subnet physically distant from the main office's DHCP server. Clients of the new subnet complain that they are not receiving IP addresses via DHCP.

- I. State what the problem might be in the case above.
- II. State two solutions for the problem identified.

I. Possible problem

- **DHCP broadcast messages do not cross routers.**
- The new subnet is **physically distant** and separated from the main office by a **router**.
- As a result, DHCP **DISCOVER** messages sent by clients in the new subnet **never reach the DHCP server**, so no IP addresses are assigned.

II. Two possible solutions

1. Configure a DHCP Relay Agent

- Enable **DHCP relay** (e.g., **ip helper-address**) on the router connecting the new subnet.
- The router forwards DHCP requests from the new subnet to the central DHCP server.

2. Install a Local DHCP Server

- Deploy a DHCP server within the new subnet.

- This server directly responds to client broadcasts without needing relay.
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2.A medium-sized company of 100+ employees has a network with the following setup:

- All internal devices use private IP addresses in the range 192.168.50.0/24.
- Their ISP has allocated the company a pool of 5 public IP addresses (198.51.100.1 to 198.51.100.5).
- All employees can access the internet simultaneously.

I. Is this setup using NAT or PAT? Explain your reasoning by identifying the key indicators in the scenario.

II. The three directors are complaining that their speed is slower during office time.

Discuss what setup you would need to change to allow the directors to use the internet without sacrificing their speed.

I. Is this setup using NAT or PAT?

This setup is using **PAT (Port Address Translation)**, also known as NAT Overload.

Reasoning:

- The company has **100+ employees**, all accessing the internet **simultaneously**.
- Only **5 public IP addresses** are available from the ISP.
- With basic NAT (one-to-one mapping), only **5 internal devices** could access the internet at the same time.
- Since *all* employees can access the internet concurrently, the router must be:
 - sharing public IP addresses, and
 - distinguishing sessions using **port numbers**.

This is the defining behavior of **PAT**, where:

- multiple private IPs → one or a few public IPs,

- connections are differentiated by **unique source ports**.

II. How to give the directors full speed without congestion

To ensure the three directors get better performance during office hours, the network setup should be adjusted using **dedicated public IPs and/or traffic prioritization**.

Recommended changes:

1. Assign Static NAT to the Directors

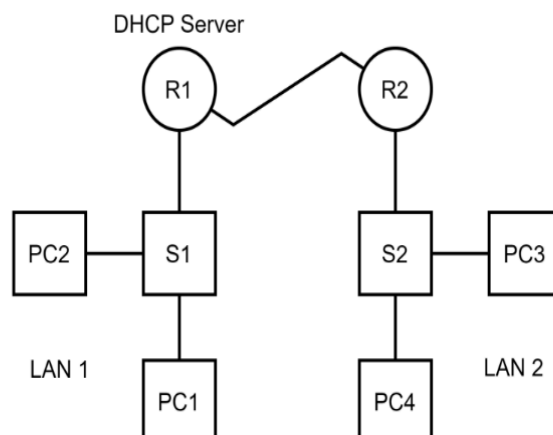
- Map **one public IP per director** using static NAT.
- This removes them from the shared PAT pool, avoiding port contention and congestion.
- Example:
 - Director 1 → 198.51.100.1
 - Director 2 → 198.51.100.2
 - Director 3 → 198.51.100.3

2. Keep PAT for Regular Employees

- The remaining employees continue sharing the remaining public IPs using PAT.

Summer 2024(A)

9.



I. No DHCP requests from any PCs of LAN2 are reaching Router R1, which has been configured as a DHCP server. **Identify** the issue and state the solution.

II. **State** the messages that are exchanged between any PC and a DHCP server for renewal of a leased IP address.

I. Issue and solution

Issue:

- **DHCP broadcasts from LAN 2 do not reach Router R1** (the DHCP server).
- This is because **DHCP uses broadcast messages**, and **broadcasts do not cross routers**.
- Router **R2** separates LAN 2 from R1, so DHCPDISCOVER/DHCPREQUEST messages from PC3 and PC4 are blocked at R2.

Solution:

- **Configure a DHCP Relay Agent on Router R2.**
 - Enable DHCP relay (e.g., **ip helper-address <R1-IP>** on R2's LAN 2 interface).
 - R2 will **forward DHCP requests as unicast** to R1.
- *(Alternative solution)* Deploy a **local DHCP server** in LAN 2.

II. Messages exchanged for renewal of a leased IP address

When a PC renews an existing DHCP lease, the following messages are exchanged:

1. **DHCPREQUEST**
 - Sent by the PC to request renewal of its current IP address.
2. **DHCPACK**
 - Sent by the DHCP server if the renewal is approved.
 - Confirms the IP address and **resets the lease timer**.

(If the server refuses, it would send a DHCPNAK, and the client must restart DHCP.)

2. Devices A and B, with IP addresses **172.16.11.5/16** and **172.16.11.11/16** respectively, both send packets using the same source port (**42230**) to a game web server on the Internet. The packets pass through an ISP router, which performs PAT using a single IP address (**209.123.123.45/24**) before forwarding them to the game web server.

I. **When** a reply from the game web server is received by the ISP router, **how** does the ISP router determine which device to send the reply, device A or B? **Explain briefly.**

II. **State** the type of address that device A and B uses, and **how** it is different from the single IP address used by the ISP router for PAT.

I. How the ISP router determines where to send the reply

1. **PAT (Port Address Translation)** modifies both the **source IP** and possibly the **source port** of outgoing packets.
2. The ISP router keeps a **PAT translation table** that maps:

Inside Local IP	Inside Local Port	Outside Global IP	Outside Global Port
172.16.11.5	42230	209.123.123.45	50001 (example)
172.16.11.11	42230	209.123.123.45	50002 (example)

Even if both devices use the same source port internally (42230), PAT assigns **different external ports** (50001, 50002) to differentiate the sessions.

- When the game server replies to 209.123.123.45:50001 or 209.123.123.45:50002, the ISP router **looks up the port in its PAT table** and forwards the packet to the correct internal device (A or B).

II. Type of address used by device A/B vs ISP router

Device	Address Type	Notes
A & B	Private IP (172.16.x.x)	Used inside the local network, not routable on the Internet.
ISP router	Public IP (209.123.123.45)	Used on the Internet to represent multiple private devices; routable globally.

Difference:

- Private IPs** are for internal communication; **cannot be used directly on the Internet**.
- Public IP** is **shared among multiple devices via PAT** to allow them to communicate with external servers.

Spring 2024 (A)

4. You are trying to access the WiFi network of a coffee shop. The WiFi router of the coffee shop sends your mobile a DHCP Offer message. **State** the source and destination MAC address and the content of this DHCP Offer. Also, **state** how your mobile will reply after receiving this offer.

Message	Source MAC	Destination MAC	Key Content
DHCP Offer	Router MAC	Your Mobile MAC	Offered IP, subnet mask, gateway, DNS, and lease time.
DHCP Request	Mobile MAC	Router MAC	Requested IP; confirms the device is joining the network.

5. Dipu was trying to communicate to a web server, where the connection is set up as shown in the figure below. The network the web server belongs to is a private network and R2 is configured to use NAT. Interpret how it will be possible for Dipu to communicate with the web server when it does not have a fixed public IP address assigned to it.



Even though the **web server is inside a private network (192.168.1.2)** and **does not have a fixed public IP**, communication is still possible because of **NAT (Network Address Translation)** configured on **R2**.

Step-by-step interpretation:

1. Dipu's PC (Client side)

- Private IP: 192.168.1.2
- Uses a random (ephemeral) port: 49153

2. R1 (Client-side NAT)

- R1 translates Dipu's private address:
192.168.1.2:49153 → 173.253.138.221:49153
- This makes Dipu reachable on the Internet using a **public IP + port**.

3. Request goes over the Internet

- Dipu sends an HTTP request to:
201.113.13.221:8080

4. R2 (Server-side NAT)

- The web server is private:
192.168.1.2:80
- R2 uses **port forwarding (static NAT / PAT)**:
201.113.13.221:8080 → 192.168.1.2:80
- So any request coming to 201.113.13.221:8080 is forwarded to the internal web server.

5. Server response

- The web server replies to Dipu.
- R2 translates the source back to 201.113.13.221:8080.
- R1 translates the destination back to Dipu's private IP.
- The response successfully reaches Dipu.